

Three-Question Practical Report

Multimodal AI, Visual Question Answering & Policy Compliance

Question 1	Policy Compliance Agent — System Design & Expected Outcome
Question 2	Image Retrieval / Visual Question Answering — Multimodal Pipeline
Question 3	Policy Compliance Agent — Evaluation Dashboard & Automated Testing

Purpose

This report consolidates three practical questions into one presentation-ready document. Question 1 is documented from the supplied experiment description. Questions 2 and 3 are supported with the supplied application screenshots and summarized as implementation, workflow, and observed-output reports.

Prepared as a college practical / experiment report

Question 1 — Policy Compliance Agent

Objective: Design an AI-powered system that automatically evaluates whether user inputs comply with predefined policies and rules, while combining deterministic rule checks with AI-based interpretation.

1. System Overview

The Policy Compliance Agent is an AI-powered application intended to evaluate text or structured user inputs against predefined organizational policies. Administrators define policies containing rules, conditions, and severity levels. During evaluation, a deterministic rule engine performs explicit checks while an AI agent interprets natural-language content and helps explain the decision.

Main workflow: Policy → Rules → User Input → Rule Evaluation → AI Analysis → Compliance Decision → Report

2. Key Features

- Policy and rule management.
- Rule-based compliance evaluation using deterministic constraints.
- AI-powered natural-language analysis and explanation.
- Synthetic data generation for safe testing without real personal information.
- Automated evaluation using accuracy, precision, recall, and F1-score.
- Compliance reports, audit logs, and human-review workflow.
- Performance dashboard for monitoring the system.

3. Decision Classes

Each evaluated case is classified as **COMPLIANT**, **NON-COMPLIANT**, or **NEEDS REVIEW**. The result can include violated rules, supporting evidence, severity, and recommended corrective action.

4. Technology Stack

Layer	Technologies
Frontend	React, TypeScript, Tailwind CSS
Backend	Python / FastAPI
Data	MySQL / SQLite
AI	LLM / Ollama
Evaluation	Scikit-learn

5. Expected Outcome

The expected outcome is an explainable and measurable compliance system. Deterministic rules provide consistent enforcement of explicit constraints, while AI interpretation supports natural-language understanding and explanations. Synthetic test cases make it possible to measure system behavior using standard classification metrics.

Result: A combined rule-engine + AI workflow that produces a traceable compliance decision and supporting report.

Question 2 — Image Retrieval / Visual Question Answering

Objective: Develop a multimodal pipeline that accepts multiple images, retrieves relevant images for a natural-language query, and performs Visual Question Answering (VQA) on the selected image.

1. Multimodal Pipeline

Pipeline: User Query → Image Processing → Multimodal Retrieval → Top Relevant Images → Gemini Vision / VQA → Final Answer

2. Demonstrated Application

The supplied application screen shows a college-lab interface titled **IMAGE RETRIEVAL + VISUAL QA**. It supports candidate image upload, sample dataset loading, retrieval, target-image selection, and a visual question-answering stage. The displayed sample dataset contains six candidate images, including a red car, dog outdoors, person with laptop and coffee, food, city skyline, and person in a red shirt.

3. Screenshot Evidence — Retrieval Interface

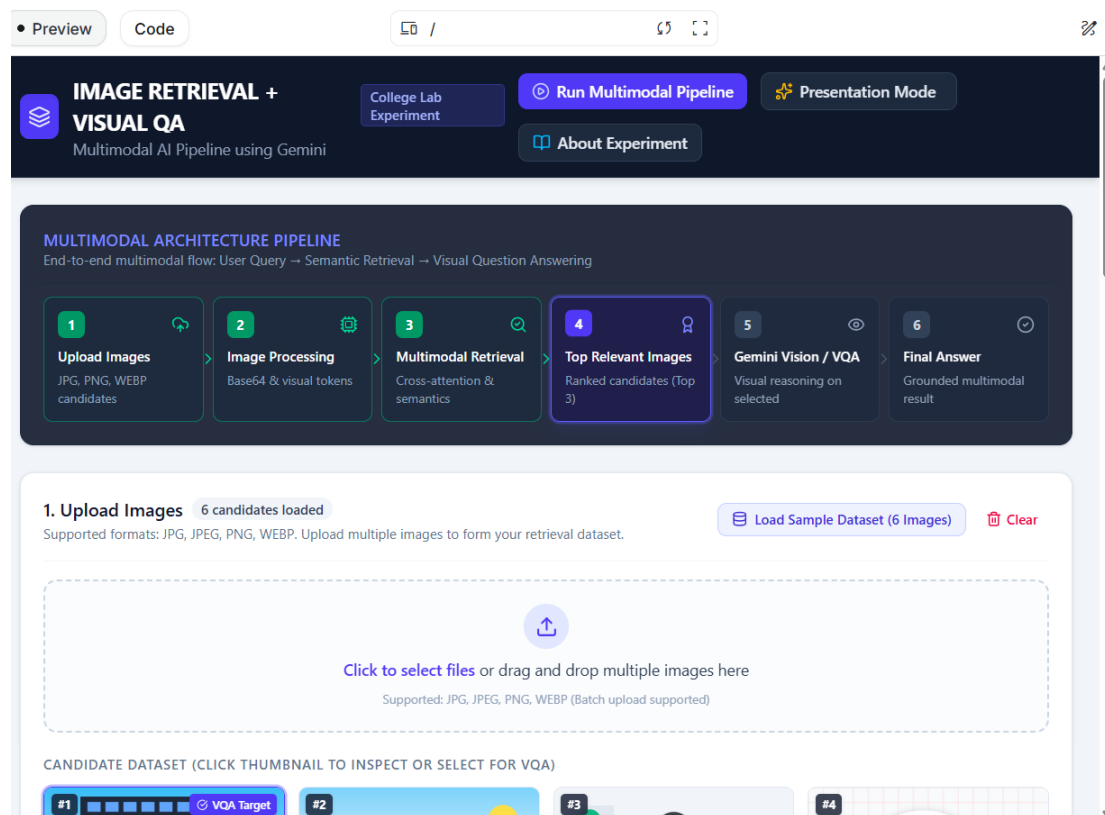


Figure 2.1 — Multimodal architecture, image upload area, candidate dataset and retrieval/VQA workflow.

4. Candidate Images and VQA Selection

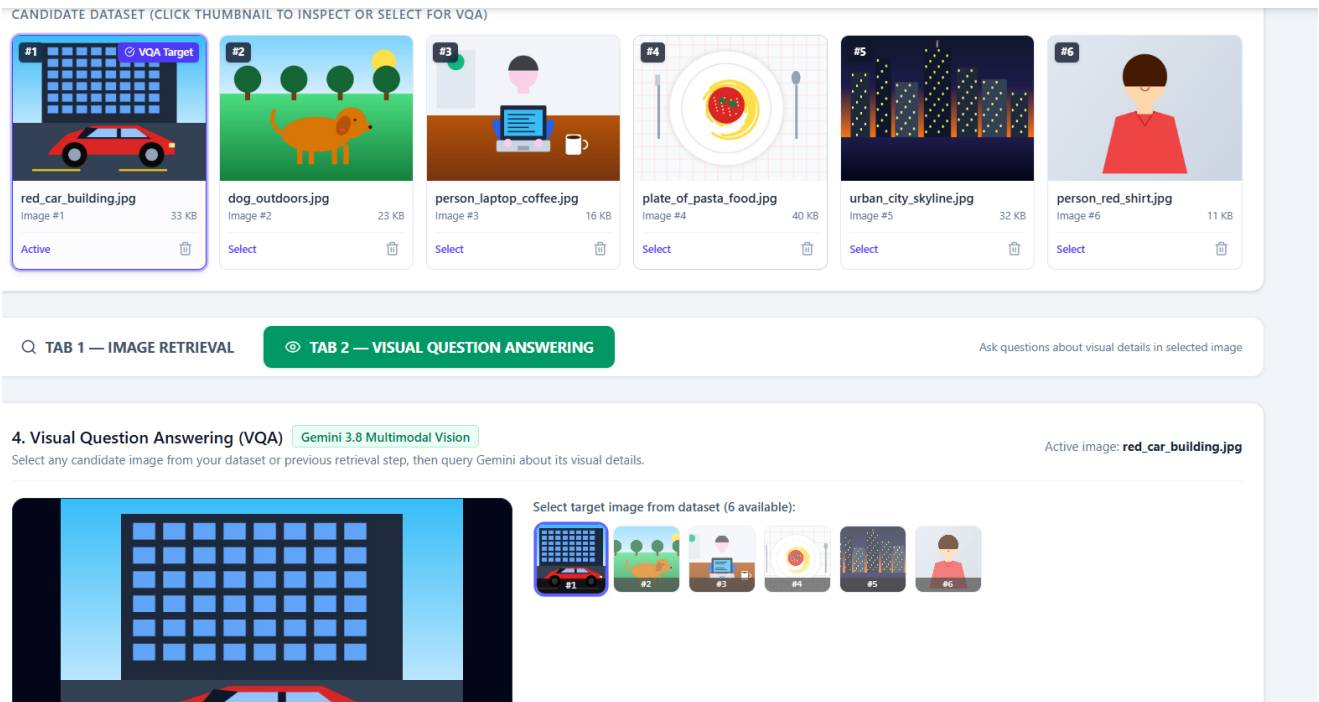


Figure 2.2 — Candidate image dataset and Visual Question Answering selection area.

The interface makes the retrieval dataset visible to the user and allows a target image to be selected for VQA. This is useful for a practical demonstration because the evaluator can see the complete path from uploaded candidates to the image chosen for visual reasoning.

5. Observed VQA Result

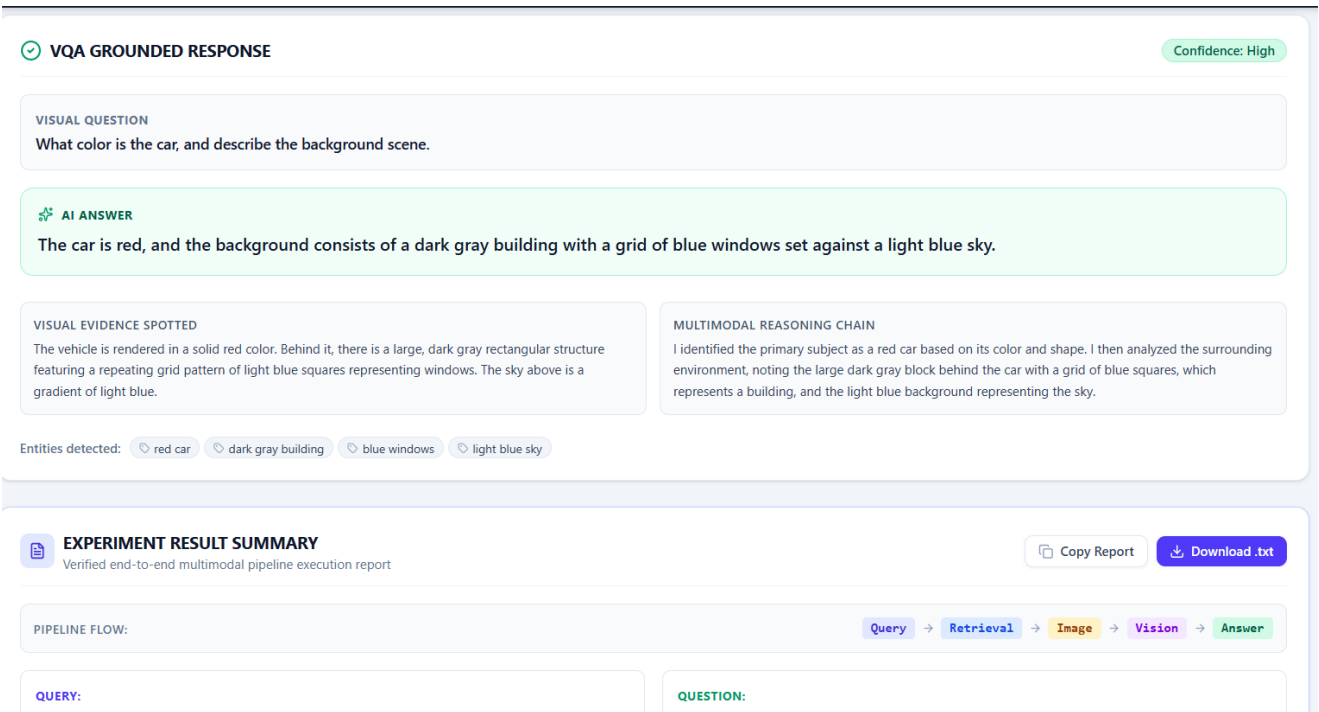


Figure 2.3 — Grounded VQA response and experiment result summary.

6. Example Demonstrated Result

Item	Observed value
Visual question	What color is the car, and describe the background scene.
AI answer	The car is red, and the background consists of a dark gray building with a grid of blue windows set against
Visual evidence	Red car; dark gray building; blue window grid; light blue sky.
Detected entities	red car; dark gray building; blue windows; light blue sky

The screenshot also presents a pipeline result summary using the sequence Query → Retrieval → Image → Vision → Answer. This demonstrates the intended end-to-end multimodal flow rather than a VQA-only interaction.

7. Expected Outcome

The experiment demonstrates how text and visual information can be combined in one pipeline: a natural-language query identifies relevant image candidates, and a multimodal model then answers a question grounded in the selected image.

Question 3 — Policy Compliance Agent: Evaluation & Monitoring

Objective: Demonstrate an enterprise-style compliance dashboard with deterministic policy enforcement, AI reasoning, synthetic test cases, automated evaluation, and measurable performance metrics.

1. Dashboard Overview

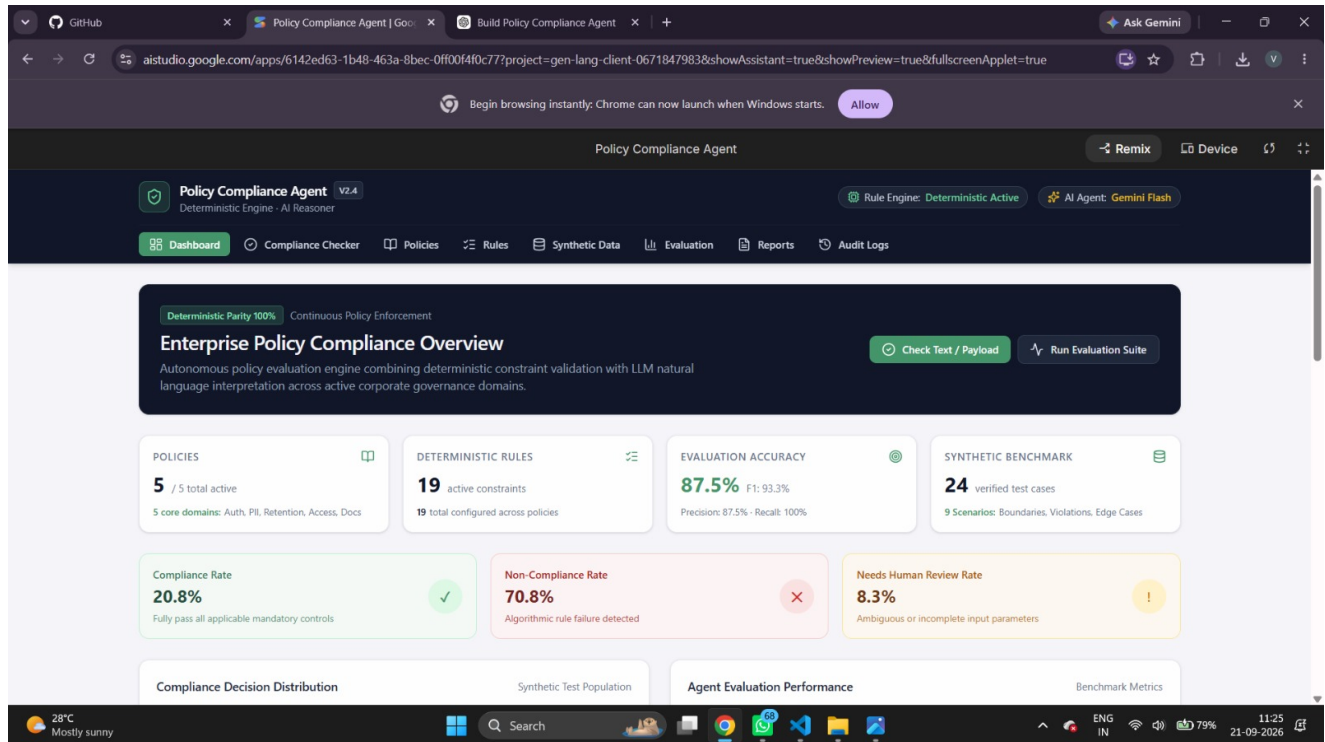


Figure 3.1 — Policy Compliance Agent dashboard overview.

The dashboard presents the system as a combination of a deterministic rule engine and an AI reasoner. It includes navigation for policies, rules, synthetic data, evaluation, reports, and audit logs. The overview also exposes compliance distribution and benchmark metrics.

2. Compliance Metrics and Decision Distribution

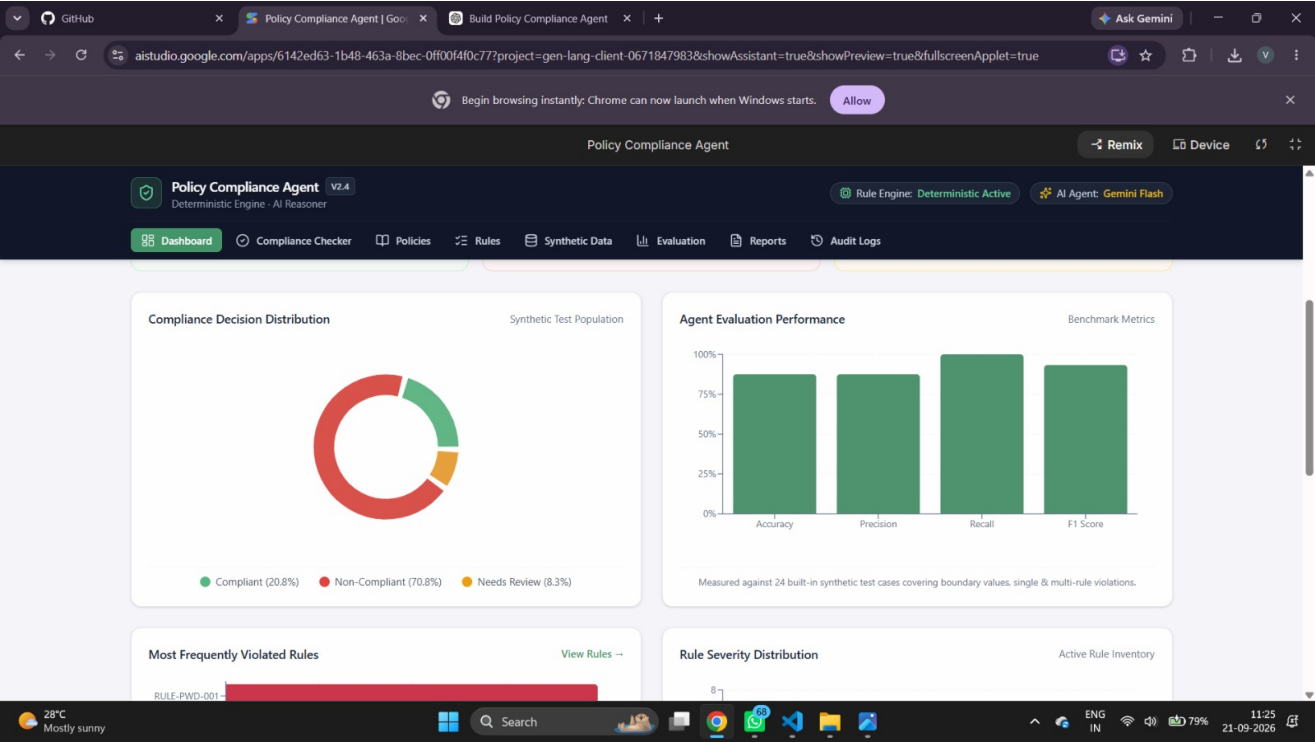


Figure 3.2 — Compliance decision distribution and agent evaluation performance.

The displayed dashboard reports a synthetic benchmark with 24 verified test cases and shows decision distribution between compliant, non-compliant, and needs-review outcomes. It also visualizes accuracy, precision, recall, and F1 score.

3. Automated Evaluation

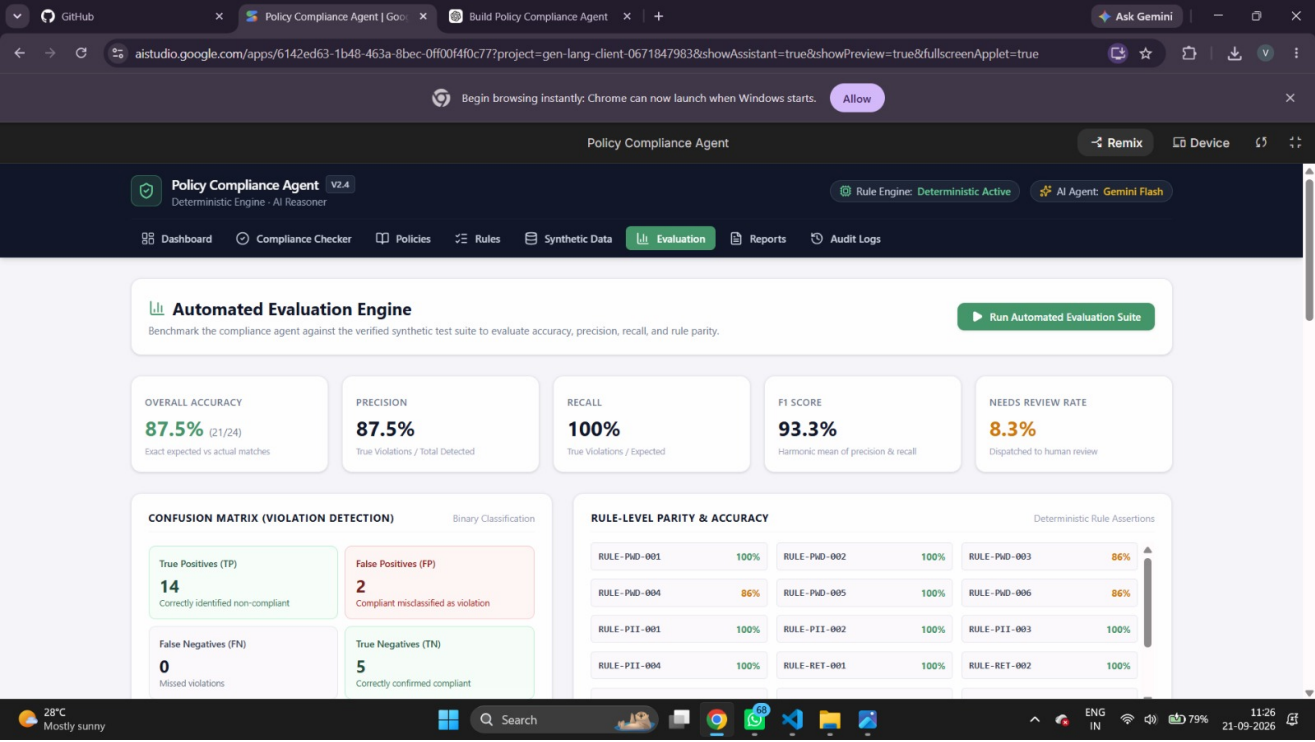


Figure 3.3 — Automated evaluation engine with classification metrics and rule-level parity.

4. Observed Evaluation Results

Metric	Displayed result	Meaning in the experiment
Overall accuracy	87.5% (21/24)	Exact expected-vs-actual matches in the displayed synthetic suite.
Precision	87.5%	Share of detected violations that were true violations in the displayed evaluation.
Recall	100%	All expected violations were detected in the displayed evaluation.
F1 score	93.3%	Harmonic mean of precision and recall.
Needs review rate	8.3%	Cases dispatched to human review in the displayed benchmark.

The evaluation screen also shows a confusion matrix for violation detection: 14 true positives, 2 false positives, 0 false negatives, and 5 true negatives. These values correspond to the displayed 24-case synthetic benchmark.

5. Key Technical Components

- Deterministic policy and rule engine for explicit constraints.
- Gemini/LLM agent for natural-language interpretation and explanations.
- Synthetic benchmark for repeatable evaluation without real personal information.
- Automated evaluation using accuracy, precision, recall and F1-score.
- Rule-level parity view to inspect individual rule behavior.
- Human-review pathway for ambiguous or incomplete cases.
- Reports and audit logs for traceability.

6. Expected Outcome

The system provides a measurable and explainable compliance workflow. The combination of deterministic rules, AI interpretation, synthetic testing, and audit-oriented reporting allows the application to demonstrate both decision-making and evaluation rather than presenting an AI answer without measurable evidence.

Note: The numerical metrics in this report are the values visible in the supplied application screenshots. They describe the displayed synthetic benchmark and should not be interpreted as production accuracy or performance on real-world data.

Overall Conclusion

Across the three questions, the demonstrated work covers two complementary AI application patterns. The multimodal experiment shows how text queries, image retrieval, and visual reasoning can be chained into an end-to-end pipeline. The policy-compliance experiments show how deterministic rules, AI interpretation, synthetic test data, and evaluation metrics can be combined into an explainable decision-support workflow.

Quick Presentation Flow

Question 1	Explain the Policy Compliance Agent architecture, workflow, features, technologies and expected outcome.
Question 2	Show the image dataset → retrieval/VQA pipeline → selected image → grounded answer.
Question 3	Show the compliance dashboard → decision distribution → automated evaluation → metrics/confusion matrix

End of Report